

TicketKN

Break-Even Analytics

How the API Calculates Event Break-Even

Client reference guide — Break-Even endpoint

API: **GET /api/users/analytics/break-even**

This document explains what data is used, how costs and revenue are combined, and how each number in the API response is calculated.

1. Overview

The Break-Even Analytics feature helps event organisers understand how many tickets must be sold to cover fixed event costs, how far they have progressed toward that goal, and what revenue/profit is projected by the event date.

Important: Fixed costs (venue, staff, marketing, equipment) are entered by the admin/organiser as query parameters each time the report is run. They are not stored permanently on the event record in the database.

2. API Request

Method: GET

URL: <https://api.ticketkn.com/api/users/analytics/break-even>

Authentication: Admin token required (x-auth-token header)

Example request

```
GET /api/users/analytics/break-even?eventId=6a36eb7486fe41ef29ee9fb8&venueCost=15&staffCost=5
```

Query parameter	Required	Description	Default
eventId	Yes	The event to analyse	—
venueCost	No*	Venue / location fixed cost	0
staffCost	No*	Staff / crew fixed cost	0
marketingCost	No*	Marketing / promotion cost	0
equipmentCost	No*	Equipment / production cost	0
variableCostPerUnit	No	Variable cost per ticket sold (v)	0
totalCosts	No	Override: single fixed-cost total	Sum of cost fields

* At least one cost input is required: either **totalCosts** or a combination of venue/marketing/staff/equipment costs totalling more than zero.

3. Data Sources

Data	Source	Used for
Event details	Event record (by eventId)	Event date, ticket plan prices (fallback)
Ticket sales	Purchase records for this event	Tickets sold, total revenue, sales dates
Fixed & variable costs	Query parameters you send	Break-even cost model

Which purchases are counted?

Only **primary ticket purchases** for the selected event are included.

- Included: standard purchases linked to the event
- Excluded: resell purchases (tickets sold on the secondary/resell market)

ticketsSold = sum of the *tickets* field on each included purchase

totalRevenue = sum of the *totalPrice* field on each included purchase

4. How Calculations Work

Step 1 — Fixed costs (F)

$F = \text{venueCost} + \text{marketingCost} + \text{staffCost} + \text{equipmentCost}$
(or use `totalCosts` directly if provided)

Example: `venueCost=15` and `staffCost=5` → **F = \$20**

Step 2 — Average ticket price (p)

If tickets have been sold:
 $p = \text{totalRevenue} \div \text{ticketsSold}$

If no tickets sold yet, the system estimates **p** from the event's configured ticket plans (weighted average of plan prices).

Step 3 — Variable cost per ticket (v)

$v = \text{variableCostPerUnit}$ from query (defaults to 0 if not sent).
Examples: printing cost per ticket, payment processing per ticket, etc.

Step 4 — Break-even ticket quantity (Q*)

$\text{Contribution margin} = p - v$
 $Q^* = F \div (p - v)$

Meaning: Q^* is the number of tickets that must be sold (at average price p) to cover all fixed costs F , after accounting for variable cost v per ticket.

Step 5 — Progress & tickets remaining

$\text{progressPercentage} = (\text{ticketsSold} \div Q^*) \times 100$
 $\text{ticketsRemaining} = \max(0, \text{ceil}(Q^* - \text{ticketsSold}))$

Step 6 — Sales velocity & trend

$\text{daysElapsed} = \text{days from first purchase (or event creation) to today (minimum 1)}$
 $\text{ticketsPerDay} = \text{ticketsSold} \div \text{daysElapsed}$

salesTrend compares ticket sales in the last 7 days vs the previous 7 days:

- **Rising** — current week sells more than 5% faster
- **Falling** — current week sells more than 5% slower
- **Stable** — otherwise

Step 7 — Projection to event date

$\text{daysUntilEvent} = \text{days from today until the event start date}$
 $\text{projectedTickets} = \text{ticketsSold} + (\text{ticketsPerDay} \times \text{daysUntilEvent})$
 $\text{projectedRevenue} = \text{projectedTickets} \times p$
 $\text{projectedProfitLoss} = \text{projectedRevenue} - F - (\text{projectedTickets} \times v)$

This is a **linear projection**: it assumes the current daily sales rate continues unchanged until the event. It does not model marketing spikes, sell-out limits, or capacity caps unless reflected in actual sales data.

Step 8 — Chart data

The API returns 21 chart points (0 to max quantity) for plotting revenue vs total cost:

- **Revenue** at quantity $Q = Q \times p$
- **Total cost** at quantity $Q = F + (Q \times v)$

Break-even occurs where revenue line crosses the total cost line (at Q^*).

5. Worked Example

Request: eventId=6a36eb7486fe41ef29ee9fb8, venueCost=15, staffCost=5
Assume the event has sold **100 tickets** generating **\$2,000** total revenue.

Metric	Calculation	Result
Fixed costs (F)	$15 + 5$	\$20
Avg ticket price (p)	$2000 \div 100$	\$20
Variable cost (v)	not provided	\$0
Break-even tickets (Q*)	$20 \div (20 - 0)$	1 ticket
Progress	$100 \div 1 \times 100$	100%
Tickets remaining	$\max(0, 1 - 100)$	0

In this example the event passed break-even after the first ticket because fixed costs were low (\$20) relative to average ticket price (\$20). Real events with higher costs or lower prices will show a higher Q*.

6. API Response Fields

Field	Description
totalCosts	Total fixed costs (F)
totalRevenue	Sum of purchase totalPrice for primary sales
ticketsSold	Total tickets sold so far
averageTicketPrice / pricePerUnit	Average revenue per ticket (p)
breakEvenTickets	Tickets needed to break even (Q*)
progressPercentage	% of break-even achieved
ticketsRemaining	Tickets still needed to reach break-even
salesTrend	Rising, Falling, or Stable
salesVelocity.ticketsPerDay	Average tickets sold per day
salesVelocity.daysElapsed	Days since sales started
projection.projectedTickets	Estimated tickets by event date
projection.projectedRevenue	Estimated revenue by event date
projection.projectedProfitLoss	Estimated profit/loss by event date
chart.points	Revenue vs cost curve for UI chart

costs.*	Echo of cost inputs used in the calculation
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7. Important Notes & Limitations

- Costs are entered manually per request — changing costs changes break-even instantly.
- Resell/secondary-market ticket revenue is excluded from this report.
- Revenue uses the owner-facing totalPrice stored on purchases (after platform fee logic in TicketKN).
- If no tickets are sold, average price is estimated from ticket plans — not from actual sales.
- Projections assume a constant daily sales rate; real-world campaigns may differ.
- The chart is a mathematical model for visualisation; actual profit depends on real sales mix and refunds.